

Summary

Dr. Saeedeh Momtazi



دانشگاه صنعتی امیرکبیر
(پلی تکنیک تهران)



آکادمی هراه



Text Processing

- Language Modeling
- Neural Networks
- Word Representation
- Sequence Modeling
- Transformers



Dialogue Systems

- Conversational agents are crucial speech and language processing applications that are already widely used commercially
- In human dialogue,
 - ▶ Speaking is a kind of action; these acts are referred to as speech acts or dialogue acts
 - ▶ Speakers also attempt to achieve common ground by acknowledging that they have understood each other
 - ▶ Conversation also is characterized by turn structure and dialogue structure



Dialogue Systems

- Chatbots are conversational agents designed to mimic the appearance of informal human conversation
 - ▶ Rule-based chatbots like ELIZA and its modern descendants use rules to map user sentences into system responses
 - ▶ Corpus-based chatbots mine logs of human conversation to learn to automatically map user sentences into system responses



Dialogue Systems

- For task-based dialogue,
 - ▶ Most commercial dialogue systems use the GUS or frame-based architecture
 - ▶ The designer specifies frames consisting of slots that the system must fill by asking the user
- The dialogue-state architecture augments the GUS frame-and-slot architecture with
 - ▶ Richer representations
 - ▶ More sophisticated algorithms for keeping track of user's dialogue acts
 - ▶ Policies for generating its own dialogue acts
 - ▶ And a natural language component



Ethical Issues in System Design

- Ethical issues have long been known to be crucial in artificial agents
- Mary Shelley's Frankenstein
 - ▶ Creating agents without a consideration of ethical and humanistic concerns
- Ethical issues:
 - ▶ **Safety:** Systems abusing users, distracting drivers, or giving bad medical advice
 - ▶ **Representational harm:** Systems demeaning particular social groups
 - ▶ **Privacy:** Information Leakage



➤ Chatbots for mental health

- ▶ Extremely important not to say the wrong thing

➤ In-vehicle conversational agents

- ▶ Must be aware of environment, driver's level of attention



Abuse and Representation Harm

- The case of Microsoft Tay
- Immediately Tay turned offensive and abusive
 - ▶ Obscene and inflammatory tweets
 - ▶ Propaganda, conspiracy theories
 - ▶ Began harassing women online
 - ▶ Reflecting racism and misogyny of Twitter users
- Microsoft took Tay down after 16 hours
- Lessons:
 - ▶ User response must be considered in the design phase
- Bias in training datasets!



Privacy: Training on user data

➤ Accidental information leakage

- ▶ "Computer, turn on the lights [answers the phone] Hi, yes, my password is..."
- ▶ Henderson show in simulation that this leakage can occur

➤ Intentional information leakage

- ▶ Dialogue systems that are designed to send user data to developer or advertisers
- ▶ Important to consider privacy-preserving dialogue systems

**Thank you for
your attention!**

